

LYRA: GENERATIVE 3D SCENE RECONSTRUCTION VIA VIDEO DIFFUSION MODEL SELF-DISTILLATION

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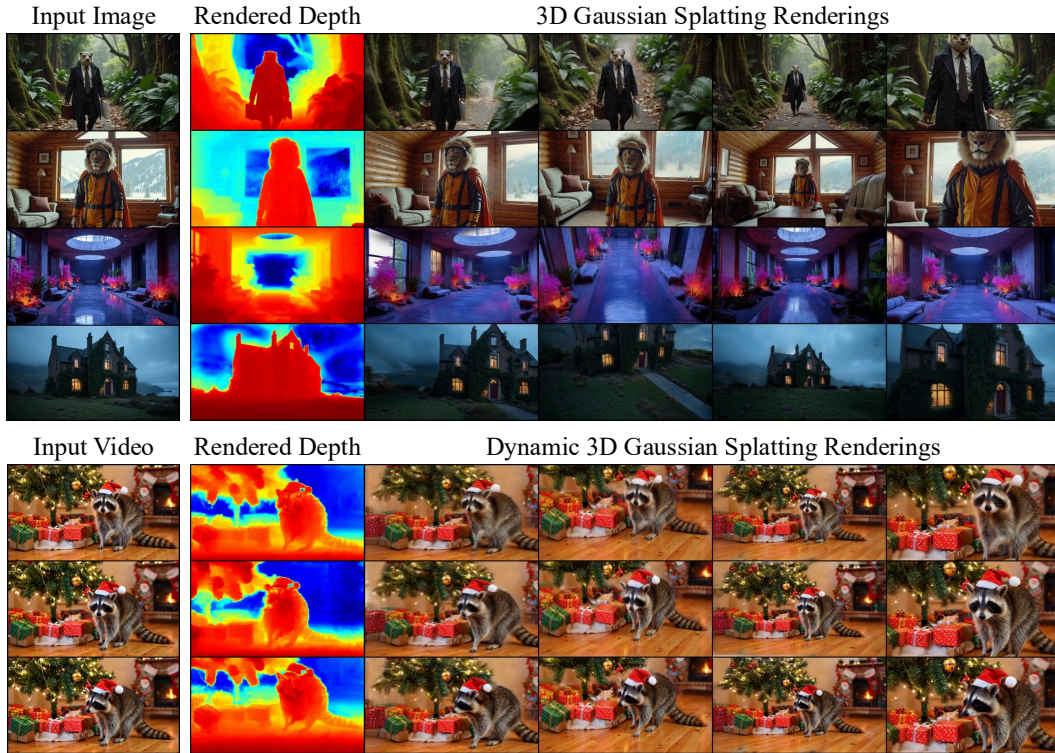


Figure 1: **Feed-Forward 3D and 4D Scene Generation.** From a single image (top), *Lyra* infers a 3D Gaussian Splatting (3DGS) representation in a feed-forward fashion, through self-distilling a video diffusion model without requiring real-world multi-view data. With a video input (bottom), *Lyra* infers a dynamic 3DGS that offers interactive control in both time (rows) and viewpoint (columns).

ABSTRACT

Explicit 3D assets are a foundational substrate for virtual environments in gaming, simulation, robotics, autonomous driving, and industrial AI, yet learning-based reconstruction systems still typically require captured real-world multi-view data, which severely constrains scale and scene diversity. Camera-controlled video diffusion models provide a complementary source of supervision: they synthesize visually rich, camera-indexed videos across broad scene distributions, but their outputs remain two-dimensional and therefore cannot directly support simulation settings that require geometrically coherent, real-time renderable worlds. We introduce a self-distillation framework that transfers the implicit 3D knowledge of such video diffusion models into an explicit 3D Gaussian Splatting (3DGS) representation without relying on multi-view training captures. The framework augments the standard RGB decoder with a trainable 3DGS decoder operating in the same latent space, and supervises the student decoder using the RGB decoder’s generated video outputs. Consequently, the 3DGS decoder is trained entirely from synthetic supervision produced by the video diffusion model while retaining an explicit representation suitable for real-time rendering. At inference time, the

resulting model synthesizes 3D scenes from either a text prompt or a single image, and the same self-distillation principle extends to dynamic 3D scene generation from a monocular input video. Across static and dynamic 3D scene generation evaluations, the framework establishes state-of-the-art performance.

1 INTRODUCTION

Scalable generation of high-quality 3D environments is a central problem in computer vision and graphics, with direct implications for film-making, VR/AR, and physical AI systems evaluated in closed-loop simulation. These settings require explicit 3D representations that render in real time, preserve multi-view consistency, and provide a coherent substrate for downstream interaction with the generated environment.

Recent neural reconstruction methods (Mildenhall et al., 2020; Kerbl et al., 2023; Moenne-Loccoz et al., 2024) can recover such representations from posed image collections, but their dependence on accurate cameras and high-quality multi-view imagery makes large-scale data acquisition a limiting factor. The limitation is more pronounced for dynamic scenes, where reconstruction often requires synchronized multi-camera capture (Li et al., 2022). Moreover, reconstruction remains bounded by observed evidence and therefore cannot reliably synthesize content beyond captured views. Feed-forward reconstruction models (Zhang et al., 2024e; Ren et al., 2024c) amortize per-scene optimization, yet they shift the bottleneck to the scarcity of diverse, large-scale 3D supervision, which restricts generalization beyond the training distribution.

Video diffusion models (Cosmos, 2025; Wan, 2025) offer a strong supervisory signal for overcoming this data limitation. Trained on large video corpora, they produce high-fidelity imagery across broad scene distributions; through exposure to real videos with handheld motion, cinematic pans, aerial trajectories, and other camera movements, they acquire implicit cues about 3D structure without explicit multi-view reconstruction datasets. Unlike purely reconstructive approaches (Charatan et al., 2024), generative video models can hallucinate plausible content outside the visible input. Their primary limitation is therefore representational rather than generative: they produce 2D frame sequences rather than explicit 3D assets, which prevents direct use in simulation and other settings requiring geometric consistency, temporal coherence, and physical interaction. Recent camera-control methods (Wang et al., 2024e; Ren et al., 2025) further demonstrate that video models can be conditioned on camera trajectories using relatively modest posed data, converting generated, camera-indexed image sequences into a practical supervisory substrate for learning explicit 3D representations.

We build on this observation by formulating *generative 3D scene reconstruction*: direct conversion of video-diffusion latents into explicit 3D environments. We introduce *Lyra*, a feed-forward framework that generates 3D Gaussian Splatting (3DGS) scenes from the latent representations of a camera-controlled video diffusion model. Its central mechanism is self-distillation (Fig. 2). Given a single image or video, *Lyra* samples a camera trajectory, denoises the corresponding video latent, and decodes that latent through two coordinated branches. The frozen RGB decoder produces the teacher video sequence, while the trainable 3DGS decoder operates in the same latent space and predicts the explicit 3D student representation. To increase viewpoint coverage, we sample multiple camera trajectories, generate multiple video latents, and train the 3DGS decoder to fuse their information into a coherent scene while reducing long-range and multi-view inconsistencies.

This design establishes three advantages that directly address the limitations of existing generative and reconstructive pipelines. **First**, it uses camera-controlled video diffusion models to synthesize large-scale and diverse supervision, eliminating dependence on captured real-world multi-view training data. **Second**, because reconstruction is performed in the compressed latent space of the video model, *Lyra* aggregates many high-resolution views without incurring the prohibitive memory cost of pixel-space feed-forward reconstruction. **Third**, the explicit 3DGS output provides a geometrically grounded representation for real-time rendering and downstream simulation. Consequently, at inference time, *Lyra* produces high-quality 3D Gaussian scenes from monocular input *without requiring any additional optimization or post-processing*.

The same self-distillation principle also extends to dynamic 4D generation from monocular video. In this setting, the video model supplies space-time supervision, and the student learns time-conditioned 3DGS representations that support novel-view synthesis of dynamic scenes.

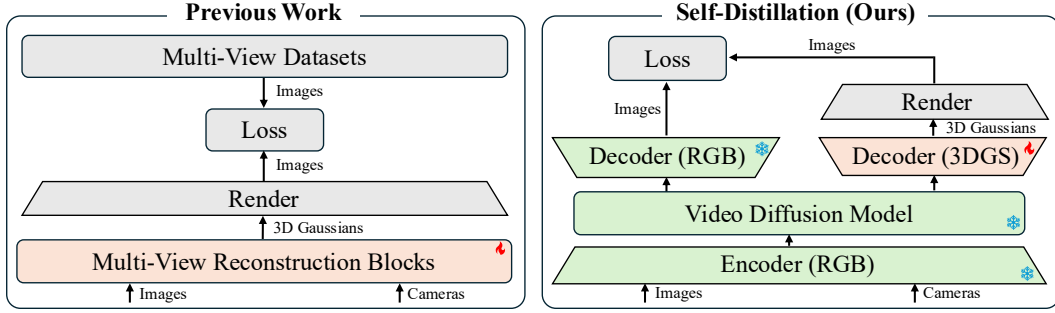


Figure 2: **Self-distillation framework of *Lyra*.** Previous work (left) (Szymanowicz et al., 2025b) trains multi-view reconstruction blocks using real-world datasets with limited diversity (Zhou et al., 2018; Ling et al., 2024b). In contrast, we propose a self-distillation framework (right) for generative 3D scene reconstruction. Precisely, a pre-trained camera-controlled video diffusion model with its RGB decoder output (teacher) supervises the rendering of the 3DGS decoder (student). Using a video model pre-trained on diverse 2D video data allows us to provide diverse multi-view supervision.

Overall, our work contributes the following:

- A self-distillation framework that trains a 3DGS student decoder from a pre-trained camera-controlled video diffusion teacher, thereby removing the need for captured real-world multi-view training data.
- A latent-space feed-forward reconstruction formulation that aggregates many generated high-resolution views while avoiding the memory limits of pixel-space reconstruction.
- A dynamic extension that applies the same formulation to feed-forward 4D reconstruction from a monocular video.
- Empirical evidence of strong generalization across diverse scenes, including state-of-the-art results in single-image 3D scene generation and single-video 4D scene generation.

2 RELATED WORK

Multi-view image generation. Early multi-view image generation methods (Watson et al., 2023; Liu et al., 2023; Shi et al., 2024; Wang & Shi, 2023) primarily address object-centric settings and typically leave scene-scale background structure under-modeled. Subsequent work (Sargent et al., 2024; Tang et al., 2023b; Schneider et al., 2025) expands the paradigm toward scene-level generation. CAT3D (Gao et al., 2024b), for example, adapts a pre-trained image diffusion model to synthesize multi-view images from generated or real image inputs, after which a separate reconstruction stage converts those views into an explicit NeRF (Mildenhall et al., 2020) or 3DGS (Kerbl et al., 2023) representation. Related space-time diffusion models extend this multi-view generation pipeline to dynamic 3D scenes observed from multiple viewpoints (Watson et al., 2024; Wu et al., 2025b; Kuang et al., 2024; Wang et al., 2025a). These approaches demonstrate the value of generative multi-view supervision, but their explicit 3D grounding still depends on an expensive per-scene optimization stage that is not amortized across generated scenes. Our work targets the complementary and more direct formulation: generating an explicit 3D scene *directly* from text or a single image.

Camera-conditioned video generation. Camera control has become a defining capability of video diffusion models. MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models with camera poses. Follow-up methods (He et al., 2024a; Xu et al., 2024c; Bahmani et al., 2025a;b) represent cameras with Plücker coordinates for pixel-wise conditioning, while another line of work (Hu et al., 2024; Hou et al., 2024; Zhang et al., 2024b) investigates training-free camera control. More recently, SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a) have shown that large-scale synthetic training data can support high-quality camera-controlled video synthesis. These systems produce camera-aware 2D sequences, but they do not by themselves yield explicit 3D structure. Our contribution is complementary: we use a camera-controlled video model as the teacher for feed-forward 3D generation, so that novel-view rendering remains available *after* generation. Concretely, we build on GEN3C (Ren et al., 2025) and distill its camera-controlled video outputs into an explicit 3DGS decoder.

Feed-forward 3D models. Early feed-forward 3D reconstruction methods largely focus on object-centric scenes without backgrounds, directly predicting NeRF (Hong et al., 2024; Li et al., 2024b) or 3DGS (Tang et al., 2024a; Szymanowicz et al., 2024) representations from text or images. Recent

scene-level systems (Charatan et al., 2024; Zhang et al., 2024e; Szymanowicz et al., 2025a; Ren et al., 2024c) typically regress per-pixel 3D Gaussians from one or more input images. Because these models are commonly trained on datasets such as RealEstate10K (Zhou et al., 2018), they can remain coupled to training-distribution scenes and may generalize poorly to diverse generated environments. Several methods therefore combine feed-forward reconstruction with generative scene synthesis. Bolt3D (Szymanowicz et al., 2025b) trains a pointmap (Wang et al., 2024b) autoencoder and generates pointmaps alongside multi-view images for feed-forward 3D reconstruction. Wonderland (Liang et al., 2025a), which is closely related to our work, combines a camera-controlled video model with a feed-forward network to generate 3D Gaussians. In contrast, we train the 3D student through self-distillation from a camera-controlled video teacher *without requiring real-world multi-view data*. We further show that the same formulation extends to feed-forward 4D scene generation, a setting not addressed by these prior systems.

3 SELF-DISTILLATION USING VIDEO DIFFUSION MODELS

Our method distills the implicit 3D knowledge encoded by video diffusion models into an explicit 3DGS decoder that produces high-quality 3D representations. We instantiate this objective as a teacher–student framework in which the video diffusion model serves as the teacher, generating RGB videos that supervise a student 3DGS decoder operating in the same latent space, as illustrated in Fig. 2. This section first specifies the camera-controlled video diffusion backbone (Sec. 3.1) and then describes how its outputs define the self-distillation signal for the 3DGS decoder (Sec. 3.2). We subsequently detail the decoder design choices in Sec. 4 and describe the minimal modifications needed to extend the pipeline to dynamic 3D scenes in Sec. 5.

3.1 BACKGROUND: CAMERA-CONTROLLED AND 3D-CONSISTENT VIDEO DIFFUSION

Because the 3DGS decoder is supervised solely by the camera-controlled video diffusion model, the quality of self-distillation depends directly on the teacher’s 3D consistency. We therefore build on GEN3C (Ren et al., 2025), a recent camera-conditioned video diffusion model designed for consistent camera-controlled generation. The following overview summarizes the GEN3C components that are used by our framework.

Spatiotemporal 3D cache. To improve video consistency and camera-control precision, GEN3C constructs a spatiotemporal 3D cache $\{\mathbf{P}^{t,v}\}$ from input image(s) or videos, where each $\mathbf{P}^{t,v}$ is a colored point cloud obtained by unprojecting the depth estimate (Wang et al., 2024a) of an RGB image captured from camera viewpoint v at time t . The cache is arranged as a $L \times V$ array, where L denotes temporal length and V denotes the number of camera views.

Rendering and structured guidance. GEN3C uses the cache by rendering each point cloud from arbitrary camera poses, producing RGB images $\mathbf{I}^{t,v}$ and disocclusion masks $\mathbf{M}^{t,v}$ via $(\mathbf{I}^{t,v}, \mathbf{M}^{t,v}) = \mathcal{R}(\mathbf{P}^{t,v}, \mathbf{C}^t)$, where $\mathcal{R}$ projects the 3D point cloud $\mathbf{P}^{t,v}$ onto the 2D camera plane according to the camera pose $\mathbf{C}^t$ at time t . Given a camera sequence $\mathbf{C} = (\mathbf{C}^1, \dots, \mathbf{C}^L)$, rendering all cache elements yields V videos of length L , which are stacked into image sequences $\mathbf{I}^v \in \mathbb{R}^{L \times 3 \times H \times W}$ and mask sequences $\mathbf{M}^v \in \mathbb{R}^{L \times 1 \times H \times W}$ for each view v . The disocclusion masks identify regions to be completed by the video diffusion model, and the rendered RGB/mask pairs provide structured visual guidance for subsequent video generation.

Video variational autoencoder (VAE). Diffusion-based video generation models commonly use a video variational autoencoder (VAE) (Kingma & Welling, 2013; Rombach et al., 2022) to compress videos into a lower-dimensional latent space for efficient training and inference. Given an RGB video $\mathbf{I} \in \mathbb{R}^{L \times 3 \times H \times W}$, the pre-trained VAE encoder $\mathcal{E}$ maps the video to $\mathbf{z} = \mathcal{E}(\mathbf{I}) \in \mathbb{R}^{L' \times C \times h \times w}$. Diffusion training and inference are then performed in this latent space, and the final video $\hat{\mathbf{I}} = \mathcal{D}_{rgb}(\mathbf{z})$ is recovered with the pre-trained VAE decoder $\mathcal{D}_{rgb}$. In this paper, we adopt the pre-trained GEN3C (Ren et al., 2025; Cosmos, 2025) model. Specifically, the latent channel dimension is $C = 16$, the temporal dimension is $L' = (L - 1)/\tau + 1$, and the spatial dimensions are $h = H/\sigma$ and $w = W/\sigma$, where the temporal compression factor is $\tau = 8$ and the spatial compression factor is $\sigma = 8$.

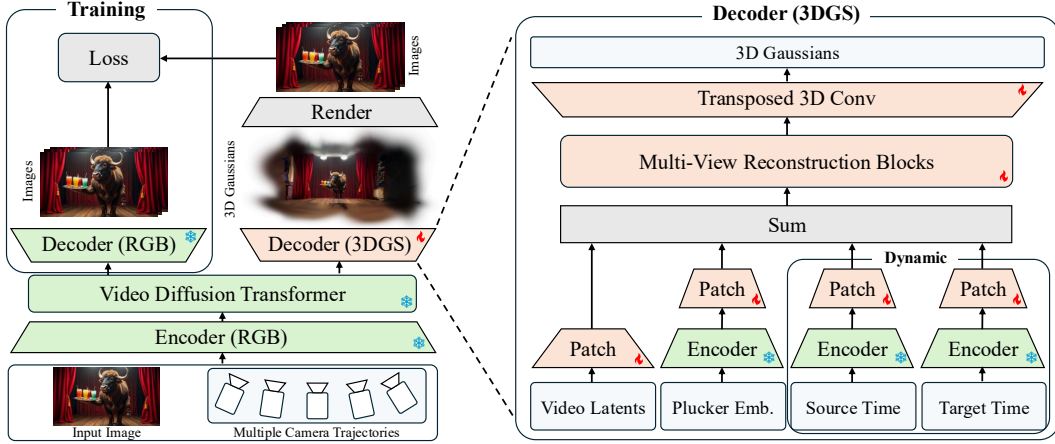


Figure 4: **3D Generative Reconstruction Framework.** Our pipeline builds upon a camera-controlled video diffusion model (Ren et al., 2025) pre-trained on large scale data. We train a 3D Gaussian Splatting (3DGS) decoder by aligning the 2D image renderings of generated 3DGS scenes with the RGB-decoded generations of the pre-trained video model. We only train the 3DGS decoder while freezing the pre-trained autoencoder and diffusion model. At inference time we directly use the 3DGS decoder, without requiring the RGB decoder anymore. Time conditioning within the 3DGS decoder allows us to easily extend our approach from static to dynamic 3D scene generation.

3.2 SELF-DISTILLATION

We train the 3DGS decoder $\mathcal{D}_s$ as a student in a teacher-student paradigm, with the camera-controlled video diffusion model $\mathcal{V}$ serving as the teacher. To obtain diverse supervision, we curate a large-scale prompt set with large language models (OpenAI, 2025; Bai et al., 2023), synthesize images I with an image diffusion model (Black Forest Labs, 2024), and expand those images into multi-view sequences using GEN3C (Ren et al., 2025).

Teacher-student setup. Given an input image I and a sampled camera trajectory $\{\mathbf{C}^t\}_{t=1}^L$, the video diffusion model $\mathcal{V}$ produces a denoised video latent $\mathbf{z} = \mathcal{V}(I, \{\mathbf{C}^t\}_{t=1}^L)$. The latent $\mathbf{z}$ is decoded by two branches: the pre-trained RGB decoder $\mathcal{D}_{rgb}$ generates teacher frames $\mathbf{I}_{\mathcal{D}_{rgb}} = \mathcal{D}_{rgb}(\mathbf{z})$, while the 3DGS decoder $\mathcal{D}_s$ predicts explicit 3D Gaussians $\mathbf{G}$. Student renderings are then defined as $\mathbf{I}_{\mathcal{D}_s} = \text{Render}(\mathbf{G}, \{\mathbf{C}^t\}_{t=1}^L)$ and supervised to match the teacher frames $\mathbf{I}_{\mathcal{D}_{rgb}}$, closing the self-distillation loop.

Multi-trajectory supervision. To enlarge viewpoint coverage around each input image, we sample $V = 6$ camera trajectories per image, as shown in Fig. 3. For each trajectory v , we construct a spatiotemporal cache $\{\mathbf{P}^{t,v}\}_{t=1}^L$ together with $L = 121$ camera poses $\{\mathbf{C}^{t,v}\}_{t=1}^L$. Passing these caches through the video diffusion model $\mathcal{V}$ yields latents $\mathbf{z}^v$, which the teacher decodes into RGB frames $\mathbf{I}_{\mathcal{D}_{rgb}}^v = \mathcal{D}_{rgb}(\mathbf{z}^v)$. The 3DGS decoder $\mathcal{D}_s$ learns to fuse the multiple $\mathbf{z}^v$ into coherent Gaussians $\mathbf{G}$ while completing disoccluded regions. This self-distillation design trains $\mathcal{D}_s$ entirely from synthetic supervision supplied by $\mathcal{V}$; operating in latent space permits efficient aggregation of multiple trajectories, while the explicit Gaussian output $\mathbf{G}$ provides geometric consistency for real-time rendering and downstream simulation.

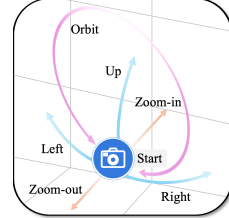


Figure 3: Sampled six camera trajectories to maximize view coverage.

4 FEED-FORWARD RECONSTRUCTION FROM MULTI-VIEW VIDEO LATENTS

The feed-forward 3DGS decoder $\mathcal{D}_s$ maps synthesized multi-view latents from the video model $\mathcal{V}$ into an explicit 3D representation that can be rendered from arbitrary viewpoints. We use 3D Gaussian Splatting (3DGS) (Kerbl et al., 2023) because it provides an explicit scene parameterization together with fast rendering. Fig. 4 (right) illustrates the decoder architecture.

4.1 3DGS DECODER

Scaling Latent-Based 3D Reconstruction. Previous feed-forward reconstruction frameworks commonly generate 3D Gaussians for each pixel, which limits the number and resolution of input views they can process. For example, GS-LRM (Zhang et al., 2024e) operates on 2–4 images at 512×512 resolution, while AnySplat (Jiang et al., 2025a) is trained with 24 images at 448×448 resolution. Our video model synthesizes $V \times L = 6 \times 121 = 726$ input views at 704×1280 resolution, a scale that exceeds the practical GPU memory capacity of pixel-space feed-forward methods. The bottleneck arises from attention over visual tokens, whose memory and computation grow with the number of pixels. We therefore avoid pixel-space scaling and instead operate directly in the compressed video latent space produced by the camera-controlled video diffusion model. We denote the resulting multi-view video latents as $\mathbf{Z} \in \mathbb{R}^{V \times L' \times C \times h \times w}$.

Architecture. The 3DGS decoder $\mathcal{D}_s$ maps its inputs into the hidden dimension of the reconstruction blocks and outputs per-pixel 3D Gaussian features $\mathbf{G} \in \mathbb{R}^{V \times L \times H \times W \times 14}$. Its inputs are the multi-view video latents $\mathbf{Z}$ and encoded Plücker embeddings $\mathbf{E}$, both represented at latent resolution as $\mathbf{Z}, \mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. Each input component is patchified to match the hidden dimension of the main reconstruction blocks. No additional visual encoder is required: a spatial 2×2 patchification layer (Dosovitskiy et al., 2021) converts the video latents into flattened reconstruction tokens. The summed inputs are processed by reconstruction blocks following Long-LRM (Ziwen et al., 2024), where each block contains one Transformer (Vaswani et al., 2017) layer followed by seven Mamba-2 (Dao & Gu, 2024) layers. Repeating this block twice yields 16 layers with 512 hidden dimensions. A transposed 3D convolution then maps the hidden representation to 14 Gaussian channels: 3D position (x, y, z) , scale (s_x, s_y, s_z) , rotation quaternion (q_w, q_x, q_y, q_z) , opacity α , and RGB (r, g, b) . Formally, the static decoder is expressed as $\mathbf{G} = \mathcal{D}_s(\mathbf{Z}, \mathbf{E})$.

Plücker embeddings. Raw Plücker embeddings are computed from camera poses $\{\mathbf{C}^{t,v}\}$ and intrinsics as $\mathbf{E}^{\text{raw}} \in \mathbb{R}^{V \times L \times 6 \times H \times W}$. We reuse the RGB encoder $\mathcal{E}$ from the pre-trained VAE to encode these embeddings. Specifically, the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins are encoded separately with $\mathcal{E}$, and the resulting latents are concatenated along the channel dimension to obtain $\mathbf{E}^{\text{enc}} \in \mathbb{R}^{V \times L' \times 2C \times h \times w}$. A lightweight MLP maps $\mathbf{E}^{\text{enc}}$ to $\mathbf{E} \in \mathbb{R}^{V \times L' \times C \times h \times w}$, reducing the channel dimension by a factor of 2 so that the camera embedding matches the video latent $\mathbf{Z}$.

4.2 LOSS FUNCTION

Image-based supervision. We supervise the 3DGS decoder with an image-based reconstruction objective comprising a Mean Squared Error (MSE) term $\mathcal{L}_{mse}$ and an LPIPS term $\mathcal{L}_{lpips}$ (Zhang et al., 2018) computed with VGG (Simonyan & Zisserman, 2014) features.

Depth supervision. A decoder trained only with RGB losses often produces flattened geometry. To regularize the recovered structure, we additionally supervise rendered depth maps using consistent video depth estimated by the off-the-shelf system ViPE (Huang et al., 2025) together with the scale-invariant depth loss $\mathcal{L}_{depth}$ from Long-LRM (Ziwen et al., 2024).

Opacity-based pruning. Following Long-LRM (Ziwen et al., 2024), we apply L1 regularization to opacity through $\mathcal{L}_{opacity}$ and remove the Gaussians with the lowest 80% opacity.

Total loss. The total objective is the weighted sum of the reconstruction, perceptual, depth, and opacity terms, with weight factors λ_i :

$$\mathcal{L} = \lambda_{mse} \mathcal{L}_{mse} + \lambda_{lpips} \mathcal{L}_{lpips} + \lambda_{depth} \mathcal{L}_{depth} + \lambda_{opacity} \mathcal{L}_{opacity} \quad (1)$$

where $\lambda_{mse} = 1.0$, $\lambda_{lpips} = 0.5$, $\lambda_{depth} = 0.05$, and $\lambda_{opacity} = 0.1$.

5 EXTENSION TO DYNAMIC 3D SCENES

The same formulation extends to dynamic 3D scenes with minimal architectural changes, as outlined below.

Self-distillation for dynamic scenes. The dynamic 3D setup follows the static formulation while replacing the single-image input with a monocular video. The video model receives a video of length

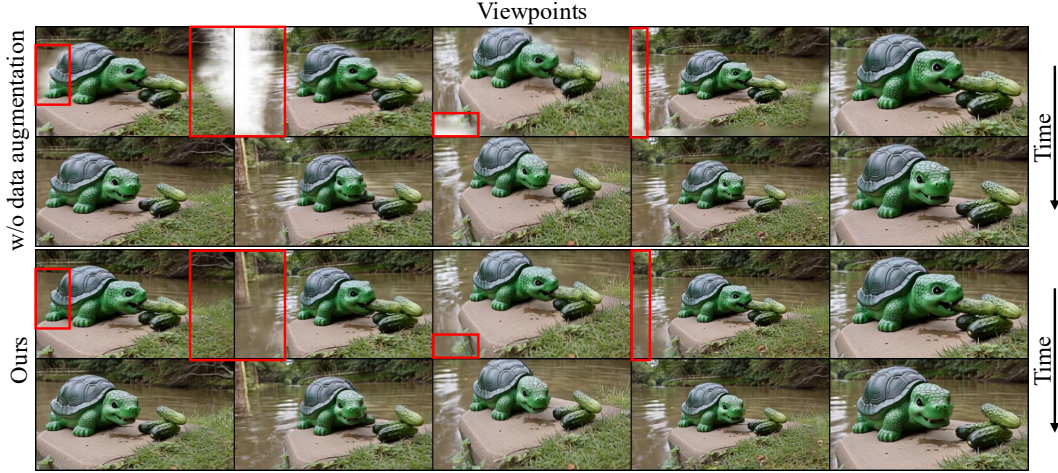


Figure 5: **Dynamic data augmentation.** When naively supervising the time-aware 3DGS decoder, we observe artifacts in the generated 3D Gaussians. Specifically, early timesteps from extreme poses exhibit low opacity in regions not covered by the supervision signal. To address this, we augment the supervision data with a motion-reversed video, ensuring that each timestep is observed from the full spatial coverage, thereby preventing low opacity artifacts in the early timesteps.

L together with corresponding camera poses and generates multi-view video latents that preserve the same underlying motion. For video inputs, we use a data-generation protocol analogous to the static case: diverse text prompts are sampled with large language models and converted into videos using video diffusion models (Cosmos, 2025; Wan, 2025). The generated videos are then annotated with camera poses and depth maps using ViPE (Huang et al., 2025).

Dynamic 3DGS decoder architecture. Following the bullet-time design of Liang et al. (2025b), we define the dynamic 3DGS decoder $\mathcal{D}_s$ to generate time-dependent 3D Gaussians for all video frames. The dynamic decoder retains the architecture of $\mathcal{D}_s$ but augments its input with encoded source and target time embeddings $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The raw source times are assigned to each input frame, forming $\mathbf{T}^{\text{src, raw}} \in \mathbb{R}^{V \times L \times 1 \times H \times W}$, while the target time is represented as $\mathbf{T}^{\text{tgt, raw}} \in \mathbb{R}^{V \times 1 \times 1 \times H \times W}$. Both tensors are spatially repeated, augmented with a 2-dimensional sinusoidal embedding to produce 3 input channels, and encoded with the RGB encoder $\mathcal{E}$ into latent tensors $\mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}} \in \mathbb{R}^{V \times L' \times C \times h \times w}$. The target time is repeated along the latent temporal dimension to match L' . During training, we randomly sample a target timestep, decode the corresponding 3D Gaussians, and supervise them across all trajectories generated with dynamic data augmentation. We fine-tune $\mathcal{D}_d$ from the pre-trained $\mathcal{D}_s$, initializing the patchification layers for $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ to zero. The time embeddings $\mathbf{T}^{\text{src}}$ and $\mathbf{T}^{\text{tgt}}$ are added directly to the sum of $\mathbf{Z}$ and $\mathbf{E}$. Formally, the dynamic decoder is defined as $\mathbf{G} = \mathcal{D}_d(\mathbf{Z}, \mathbf{E}, \mathbf{T}^{\text{src}}, \mathbf{T}^{\text{tgt}})$.

Dynamic data augmentation. In static 3D scenes, where time is not a variable, frames from different timesteps can all supervise the same 3DGS rendering. In dynamic 3D scenes, however, the 3DGS varies over time, so only frames at the corresponding timestamp provide valid supervision. This restriction can induce a degenerate solution in which the 3DGS at a given timestep ignores information from frames at other timesteps (see Fig. 5).

To mitigate this issue, we introduce a dynamic data augmentation strategy that balances supervision across timesteps. Early frames are naturally associated with viewpoints close to the input image, whereas later frames correspond to more distant viewpoints. We counter this imbalance by augmenting the training data with paired supervision views for each timestep, one near and one far. Concretely, we reverse the input video in frame order and pass it through the video model $\mathcal{V}$, which produces six additional multi-view sequences. After reversing these sequences back to the original motion order, we obtain six trajectories that progress inward, from far viewpoints toward the input. Combined with the original six outward trajectories, this yields 12 supervision views per timestep. This augmentation is applied only during training: the reversed trajectories provide additional supervision that prevents collapse from pruning artifacts and are not required at inference time.

Table 1: **State-of-the-art comparisons.** We compare our method with previous works for single image to 3D generation using RealEstate10K, DL3DV, and Tanks-and-Temples datasets.

Method	RealEstate10K			DL3DV			Tanks-and-Temples		
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
ZeroNVS	13.01	0.378	0.448	13.35	0.339	0.465	12.94	0.325	0.470
ViewCrafter	16.84	0.514	0.341	15.53	0.525	0.352	14.93	0.483	0.384
Wonderland	17.15	0.550	0.292	16.64	0.574	0.325	15.90	0.510	0.344
Bolt3D	21.54	0.747	0.234	-	-	-	-	-	-
Ours	21.79	0.752	0.219	20.09	0.583	0.313	19.24	0.570	0.336



Figure 6: **Image-to-3DGS Generation.** We visualize five views from generated 3DGS scenes.

6 EXPERIMENTS

6.1 EXPERIMENTAL SETUP

Datasets. To train our model, we do not use any existing multi-view datasets; instead, we construct our own, which we call the *Lyra* dataset, relying solely on our video model to supervise the 3DGS decoder. The 3D reconstruction setup uses 59,031 images, while the 4D setup has 7,378 videos. All data are from diverse text prompts, spanning scenarios such as indoor and outdoor environments, humans, animals, and both realistic and imaginative content. We synthesize six camera trajectories for each image (3D) or video (4D), yielding 354,186 videos for 3D and 44,268 videos for 4D.

Baselines. We compare our method with the state-of-the-art approaches for generative single-image-to-3D reconstruction, i.e., ZeroNVS (Sargent et al., 2024), ViewCrafter (Yu et al., 2024b), Wonderland (Liang et al., 2025a), and Bolt3D (Szymanowicz et al., 2025b). Note that *no source code is available* to evaluate these methods on our out-of-distribution evaluation set, hence we mainly rely on reported quantitative comparisons from the papers.

Evaluation. We follow previous works to evaluate our model on the task of single-image to 3D using RealEstate10K (Zhou et al., 2018), DL3DV (Ling et al., 2024b), and Tanks and Temples (Knapitsch et al., 2017). We follow the evaluation protocol outlined in Wonderland (Liang et al., 2025a) and Bolt3D (Szymanowicz et al., 2025b). We evaluate the performance using standard reconstruction metrics, i.e., PSNR, SSIM, and LPIPS.

6.2 MAIN RESULTS

Quantitative results. We show quantitative evaluations for single image-to-3D in Tab. 1. Our method outperforms previous works on all benchmarks across all metrics. Hence, the main improvements to further boost the quality lie in the development of stronger video generative models, as our reconstructions will directly benefit from them. We provide additional quantitative comparisons on the *Lyra* dataset in Appendix C.



Figure 7: **Ablations.** We visualize ablation results by rendering the same extreme novel viewpoint after image-to-3DGS generation; depth visualizations and ablations are provided in Appendix C.4.

Qualitative results. We visualize novel view renderings from our generated 3DGS scenes in Fig. 1 and Fig. 6; video results will be released upon acceptance. Our method generates high-quality novel view content for 3D/4D scenes in unseen regions while maintaining consistency with the input image/video. We visualize qualitative comparisons on the *Lyra* dataset in Appendix C.

6.3 ABLATIONS

In Tab. 2, we motivate our design choices by ablating key components on out-of-distribution diverse prompts. We compare 3DGS renderings with videos generated by our camera-controlled video diffusion model.

Real data only. Instead of using self-distillation, we train a model using only real multi-view datasets, i.e., RealEstate10K (Zhou et al., 2018) and DL3DV (Ling et al., 2024b). These datasets are commonly used in previous works, but lead to limited generalizability to out-of-distribution scenes.

Self-distillation + real data. Perhaps surprisingly, joint training with self-distillation and real data does not improve over using only self-distillation without real data. This confirms that self-distillation is diverse and consistent enough to learn a reconstruction model.

No depth loss. Depth loss prevents flat geometry and even slightly improves image-based metrics. We visualize depth maps in Appendix C.4.

No opacity pruning. We keep all Gaussians instead of pruning them by opacity. Learning to prune most of the 3D Gaussians makes the output representation more compact and slightly improves visual quality. Rendering at $H = 704$, and $W = 1280$ takes 18ms with pruning vs. 30ms without pruning, a $1.67\times$ speed up.

No LPIPS loss. Adding the LPIPS loss improves the robustness to input inconsistencies and enhances the preservation of high-frequency details.

No multi-view fusion. A naive implementation would be to generate 3D Gaussians for each camera trajectory independently and then fuse the 3D Gaussians into one point cloud. However, we observe significant advantages by learning the multi-view fusion from different trajectories with our reconstruction blocks, where each token attends to the others.

No Mamba-2. We replace our joint Transformer/Mamba-2 with Transformer-only blocks and observe slightly lower quality. One forward pass with $V = 6$, $L = 121$, $H = 704$, and $W = 1280$ takes 3213ms with joint blocks in comparison to 20922ms with Transformer-only blocks, a $6.5\times$ speedup.

No latent-based 3DGS. Previous works operating in the pixel space, such as BTimer (Liang et al., 2025b), are restricted to 12 input frames, while we can take up to 726 input frames. Consequently, operating the 3DGS decoder in the pixel space instead of the latent space leads to out-of-memory.

Table 2: **Ablation study on *Lyra* dataset.**

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
Ours	24.77	0.837	0.224
Data			
real data only	19.08	0.659	0.413
self-distill. + real data	24.74	0.823	0.236
Loss			
w/o depth loss	24.31	0.811	0.247
w/o opacity pruning	24.55	0.820	0.237
w/o LPIPS loss	23.74	0.766	0.370
Architecture			
w/o multi-view fusion	17.73	0.632	0.446
w/o Mamba-2	24.58	0.818	0.241
w/o latent 3DGS		OOM	

7 CONCLUSION

In this work, we propose *Lyra*, a novel 3D and 4D generation framework relying only on a single image or video input. Instead of collecting multi-view datasets, we introduce camera-controlled video diffusion models as teachers within a self-distillation framework for a student 3DGS decoder. Our 3DGS decoder operates in the latent space of the video model and directly reconstructs 3D Gaussians without any post-processing or optimization. This design enhances generalizability and coverage across diverse scenes, while our dynamic extension demonstrates the feasibility of 4D generation from monocular video. Currently, the scale and consistency of our generated scenes are bounded by the capacity of our camera-controlled video diffusion model. Hence, further improving the video diffusion model will enable large-scale, consistent scene synthesis. Moreover, investigating the adaptation of auto-regressive techniques (Chen et al., 2024a) into our framework for large-scale generation is an interesting direction for future work. Lastly, modelling motion and tracking information within the reconstruction network, as in concurrent work (Lin et al., 2025), could improve visual motion quality.

8 ETHICS STATEMENT

Like all generative AI technologies, there are risks of misuse of our method, including producing misleading or synthetic 3D/4D content. While such risks exist, the main goal of this research is to contribute to the fields of simulation, robotics, and embodied AI by providing scalable and controllable tools for data generation. In this sense, we expect our most significant contributions to be made through the following capabilities and insights.

- **The ability to enable realistic simulation for embodied agents:** We provide controllable and physically consistent camera motion within generated 3D and 4D environments to support training and evaluating agents that must perceive, navigate, and interact with complex scenes.
- **Improved scalability for data engines:** Our method removes the need for multi-view capture and expensive per-scene optimization, and helps produce high-fidelity, customizable scenes at scale, which is particularly suitable for research in robotics, reinforcement learning, and closed-loop simulation.
- **Better technical understanding:** As an academic study, this work contributes a foundational case study of how camera control interacts with spatiotemporal generative models, which will benefit research in computer vision, graphics, and embodied AI.

While we caution against deceptive or wrongful use of this technology—where misleading and/or unverifiable media could be created with realistic generative outputs—we believe that the adoption of safeguards (e.g., provenance tracking, dataset documentation, and good evaluation practices) will be useful in ensuring that generative models advance science and society in positive ways.

9 REPRODUCIBILITY STATEMENT

To fully reproduce our results and accelerate future research in this area, our training and inference code, model weights, and data for both 3D and 4D generation will be released upon acceptance.

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A ADDITIONAL DETAILS OF VIDEO DIFFUSION MODEL

This section provides additional information on the GEN3C (Ren et al., 2025) video diffusion model backbone introduced in Sec. 3.1 and how we further improve its 3D consistency.

Conservative mask refinement. A key limitation of GEN3C’s forward warping is background leakage, where occluded object parts in the source view (e.g., lion body in Fig. 8) become visible in novel viewpoints but are not properly indicated in the disocclusion mask. This leads to incomplete foreground completion by the video diffusion model, as shown in Fig. 8 (second and fourth columns).

To address this, we adopt a conservative strategy: since the geometry of occluded regions in $\mathbf{P}^{t,v}$ is unknown, we assume all such areas are occupied and derive refined disocclusion masks accordingly. This allows the video model to reason about potentially visible foreground regions that were missed by standard point-based rendering.

Our approach constructs a triangular mesh from camera-space points $\mathbf{P}_c^{t,v} = \mathbf{C}^t \mathbf{P}^{t,v}$ by connecting spatially adjacent pixels, similar to Hu et al. (2021):

$$\mathcal{M} = \bigcup_{(u,v)} \{ \triangle(\mathbf{p}_{u,v}, \mathbf{p}_{u+1,v}, \mathbf{p}_{u,v+1}), \triangle(\mathbf{p}_{u+1,v}, \mathbf{p}_{u+1,v+1}, \mathbf{p}_{u,v+1}) \} \quad (2)$$

where $\mathbf{p}_{u,v}$ denotes the 3D point at pixel (u, v) in $\mathbf{P}_c^{t,v}$. This surface mesh serves as a boundary between visible and invisible regions. For each rendered pixel, we compare the standard point-based depth $\mathbf{D}^{t,v}$ with the mesh-interpolated depth $\mathbf{D}_{\mathcal{M}}^{t,v}$ obtained via ray-surface intersection:

$$\mathbf{M}^{t,v}(u, v) = \begin{cases} 0 & \text{if } \mathbf{D}_{\mathcal{M}}^{t,v}(u, v) < \mathbf{D}^{t,v}(u, v) - \epsilon \\ \mathbf{M}_{\text{orig}}^{t,v}(u, v) & \text{otherwise} \end{cases}, \quad (3)$$

where $\mathbf{M}_{\text{orig}}^{t,v}$ denotes the original disocclusion mask from forward warping and ϵ is a small tolerance factor. When the mesh surface is closer than the original points, we conservatively mask out these regions as potential foreground disocclusions. This produces more reliable guidance for video generation, especially under large viewpoint changes, as shown in Fig. 8 (third and fifth columns).

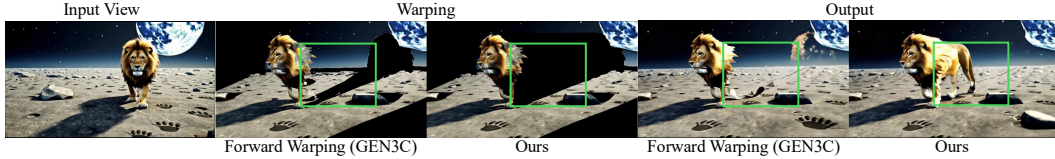


Figure 8: **Rendering function comparison.** Our model builds upon GEN3C (Ren et al., 2025) that uses forward warped images as camera control conditioning. We visualize forward warping (left) vs. our conservative rendering function (right). The highlighted region is incorrectly filled with background pixels in forward warping but properly masked in ours, enabling correct completion.

B MODEL DETAILS

Progressive training. Directly training the 3DGS decoder on 6 trajectories of 121 frames at 704×1280 is expensive, hence we train our network progressively, as shown in Tab. 3. Furthermore, we use the final static pre-trained model to initialize our dynamic model. The total training time takes 6 days with 8 NVIDIA A100 80 GB GPUs. We use gsplat (Ye et al., 2025) as the 3DGS implementation.

Subsampling Gaussians. Generating pixel-aligned 3D Gaussians scales linearly with each spatial and temporal resolution. Our generated scenes represent 726 frames at 704×1280 resolution, leading to 654,213,120 per-pixel Gaussians. Hence, we only generate one 3D Gaussian per 8×8 spatial neighborhood, reducing the total number of 3D Gaussians by a factor of 64, i.e., 10,222,080. Moreover, our opacity-based pruning further reduces the number to 2,044,416, making the output more compact.

Joint Transformer-Mamba blocks. We use the reconstruction block design introduced in Long-LRM (Ziwen et al., 2024), i.e., one block consists of one Transformer layer followed by seven

Table 3: **Progressive training setup.**

Stage	$H \times W$	L	V	S	B	Steps
Static						
1	176×320	17	1	17	4	10k
2	176×320	49	1	49	4	2.5k
3	352×640	49	1	49	2	2.5k
4	704×1280	49	1	49	1	2.5k
5	704×1280	121	1	9	1	57.5k
6	704×1280	121	1-6	9	1	7k
Dynamic						
7	704×1280	121	6	12	1	10k

Mamba-2 layers. We repeat the block twice to get 16 layers with 512 hidden dimensions. Note that our network is significantly smaller than the one used in Long-LRM (Ziwen et al., 2024) or Wonderland (Liang et al., 2025a), which use 24 layers with 1024 hidden dimensions. While pure Transformers are typically more powerful, they are significantly slower in training and inference than Mamba2. Since we trained both configurations within the same training budget, we hypothesize that transformer-only blocks would need considerably more training iterations to fully catch up in rendering quality.

Camera encoding. We use Plücker embeddings for camera encoding, following prior works (Kant et al., 2024; Zhang et al., 2024e). The embeddings are computed at full spatial and temporal resolution and then projected into the video latent space using the pre-trained RGB encoder. Since the RGB encoder expects 3 input channels but Plücker embeddings are 6-dimensional, we separately encode the 3-dimensional ray directions and the 3-dimensional cross product of ray directions and origins. The resulting latent encodings are then concatenated along the channel dimension.

Time encoding. Similarly, to make the 1-dimensional time input compatible with the 3-channel input expected by the RGB encoder, we concatenate a 2-dimensional sinusoidal embedding to the original time value along the channel dimension. We then replicate these values across the spatial dimensions before encoding the time into the video latent space. This approach allows us to independently encode both the source and target times into the compressed latent representation.

Dynamic data augmentation. We visualize our dynamic data augmentation procedure for two example camera trajectories in Fig. 9. The procedure creates pairs of supervision views that cover the scene from the same motion state but different extreme viewpoints.

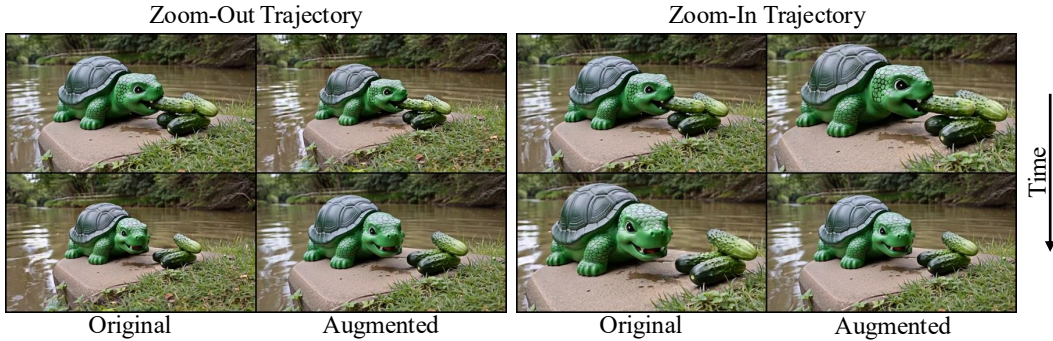


Figure 9: **Dynamic data augmentation videos.** We augment the supervision data with a motion-reversed video, ensuring that each timestep is observed from the full spatial coverage, thereby preventing low opacity artifacts in the early timesteps. We show two example trajectories, i.e., zoom-out and zoom-in, and visualize their corresponding augmented videos. The augmented videos are flipped in their camera motion.



Figure 10: **Image-to-3DGS Comparison.** We compare our method with BTimmer (GEN3C) and visualize five views from generated scenes. We observe significantly fewer artifacts and higher fidelity.

C ADDITIONAL EVALUATION

C.1 ADDITIONAL BASELINE DETAILS

We additionally compare our model against BTimmer (Liang et al., 2025b), a recent approach for joint 3D and 4D reconstruction from posed images or videos. We reached out to the authors of BTimmer to conduct experiments for us using the same scenes as *Lyra*. Since BTimmer is purely regression-based, we integrate it with our GEN3C (Ren et al., 2025) video diffusion backbone. Specifically, we use the same camera trajectories as in our model and provide RGB-decoded videos from GEN3C as input to BTimmer. BTimmer operates in pixel space and requires 12 input images. However, as discussed in the main paper, pixel-space models such as BTimmer run out of memory when directly using the 726 high-resolution frames generated by GEN3C. To address this, we uniformly subsample GEN3C frames. For static evaluations, we select 12 frames that evenly cover the available viewpoints. For dynamic evaluations, we sample 11 frames to span the viewpoint range and additionally include the bullet-time frame corresponding to the motion being reconstructed. This ensures that the target motion state is always present as a reference. We denote this integrated baseline as BTimmer (GEN3C), i.e., BTimmer combined with GEN3C.

C.2 ADDITIONAL STATIC 3D EVALUATION

For the task of image to 3DGS generation, we show qualitative comparisons with BTimmer (GEN3C) in Fig. 10 and qualitative results in Tab. 4a. We observe higher quality and less artifacts for our method.

C.3 DYNAMIC 3D EVALUATION

BTimmer (GEN3C) comparison. For dynamic 3D evaluation, we compare our method with BTimmer (GEN3C), as discussed in Sec. C.1, on 100 out-of-distribution videos from our dynamic *Lyra* dataset. We crop the video to 512×512 for fair comparisons, as BTimmer was mainly trained on that resolution. We show results in Tab. 4b and observe that our method significantly outperforms BTimmer (GEN3C).

Table 4: Quantitative results on static and dynamic *Lyra* datasets.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$
BTimer (GEN3C)	16.32	0.580	0.427	BTimer (GEN3C)	20.29	0.687	0.315
Ours	24.92	0.834	0.183	Ours	23.07	0.779	0.231

(a) Comparison on static *Lyra* dataset.

(b) Comparison on dynamic *Lyra* dataset.

C.4 DEPTH VISUALIZATION

We visualize depths from our generated 3DGS in Fig. 11. Using depth supervision prevents flat geometries without sacrificing visual quality.

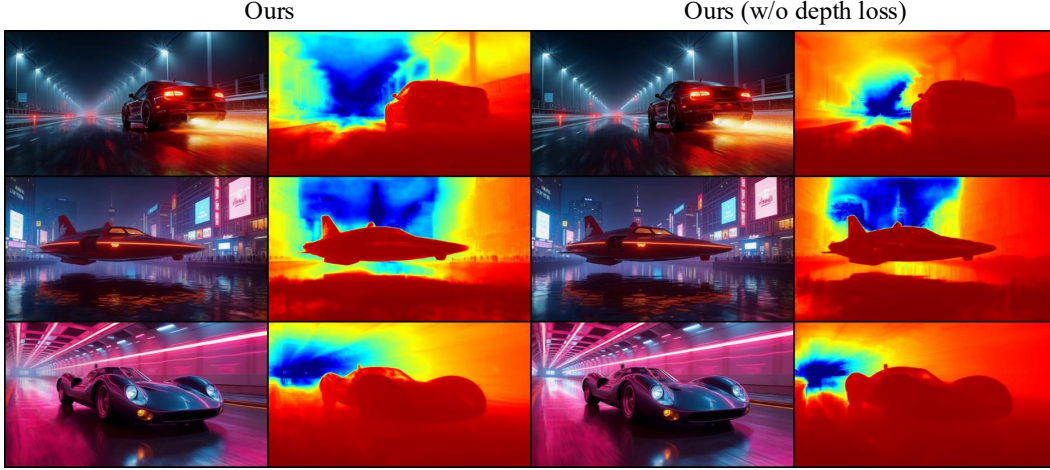


Figure 11: **Depth loss ablation.** We visualize 3DGS renderings and corresponding depth maps. Using the depth loss as additional supervision prevents flat geometries without sacrificing visual quality.

D ROBOT SIMULATION

One of our primary motivations for building a diverse 3D generator is its ability to synthesize simulation environments for autonomous agents. To this end, we develop an end-to-end pipeline: we first generate 3D scenes from text with our model, export the 3D Gaussians as a .ply file, and convert the .ply files into a .usdz format. We use 3DGUT (Wu et al., 2025a) and its export function to create .usdz files. The resulting .usdz file can then be imported into the NVIDIA Isaac robot development platform as a physically based virtual environment. AI-based robots can be trained and tested under diverse conditions within these environments. Early demo results will be released upon acceptance; we include a screenshot of the interface in Fig. 12.

E RELATED WORK

In this section, we will discuss additional related work. We highlight the differences to previous works CAT3D (Gao et al., 2024b) and Bolt3D (Szymanowicz et al., 2025b) in Fig. 13.

3D generation. Early approaches to 3D generation primarily focused on single object categories, extending GANs into the 3D domain by leveraging neural renderers as an inductive bias (DeVries et al., 2021; Chan et al., 2022; Or-El et al., 2022; Schwarz et al., 2022; Bahmani et al., 2023b). With the introduction of CLIP-based supervision (Radford et al., 2021), the field advanced toward more flexible and diverse asset creation, enabling both text-driven generation and editing (Chen et al., 2018; Jain et al., 2022; Sanghi et al., 2022; Jetchev, 2021; Gao et al., 2023; Wang et al., 2022). More recently, diffusion models have significantly improved quality by replacing CLIP guidance with Score Distillation Sampling (SDS) (Poole et al., 2023; Wang et al., 2023b; Lin et al., 2023a; Chen et al.,

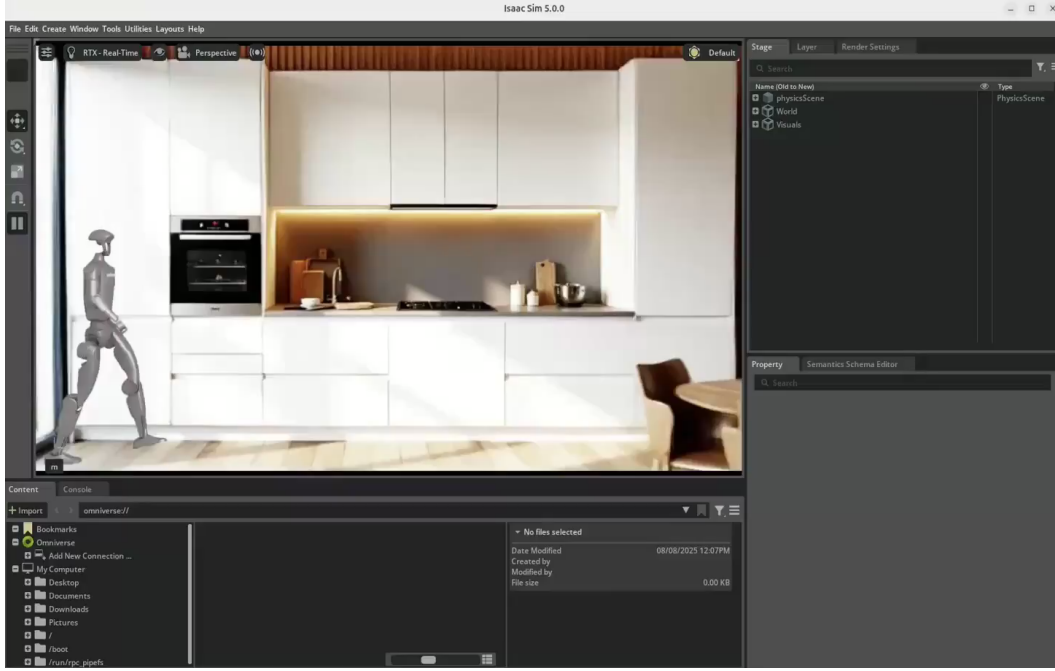


Figure 12: **Robot Simulation.** We visualize a frame of a robot simulation within the Isaac Sim 5.0 simulation framework that takes a generated 3DGS scene from our method as input.

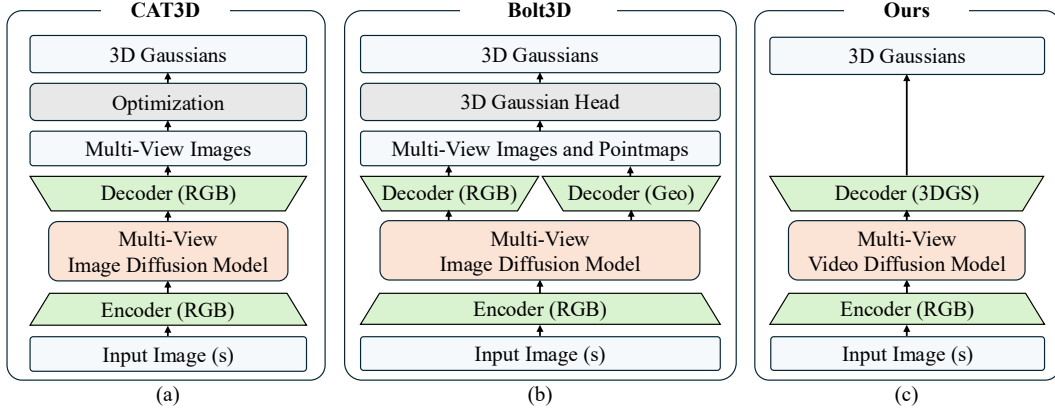


Figure 13: **Approaches for 3D Generation.** CAT3D (Gao et al., 2024b) proposed a multi-view image diffusion model that outputs images from novel viewpoints; subsequently, the images are reconstructed into 3D Gaussians using optimization. Bolt3D (Szymanowicz et al., 2025b) fine-tunes CAT3D to output pointmaps using a geometry autoencoder; instead of relying on optimization, the 3D scene is predicted with a feed-forward 3D Gaussian head. In contrast, our work builds upon a pre-trained camera-controlled video diffusion model and *directly* decodes the multi-view video latents into 3D Gaussians.

2023b; Liang et al., 2023; Wang et al., 2023a; Li et al., 2024e; He et al., 2024b; Ye et al., 2024; Liu et al., 2024b; Yu et al., 2023; Katzir et al., 2024; Lee et al., 2024a; Sun et al., 2024a).

To further enhance structural coherence, several methods enforce multi-view consistency by generating scenes from different perspectives (Lin et al., 2023b; Liu et al., 2023; Shi et al., 2024; Feng et al., 2024a; Liu et al., 2024a; Kim et al., 2023; Voleti et al., 2024; Höllein et al., 2024; Tang et al., 2024c; Gao et al., 2024b; Wang et al., 2025f; Kant et al., 2025; Yuan et al., 2025; Ren et al., 2024b), while others adopt iterative inpainting as a strategy for scene synthesis (Höllein et al., 2023; Shriram et al., 2024). Another line of work explores lifting 2D images into 3D using NeRF (Mildenhall et al., 2020), 3D Gaussian Splatting (Kerbl et al., 2023), or mesh-based representations coupled with diffusion

models (Chan et al., 2023; Tang et al., 2023a; Gu et al., 2023; Liu et al., 2024c; Yoo et al., 2023; Tewari et al., 2023; Qian et al., 2024b; Long et al., 2024; Wan et al., 2024; Szymanowicz et al., 2023; Lu et al., 2024).

Feed-forward 3D models. Complementary efforts investigate fast, feed-forward techniques that directly predict 3D content from images or text inputs (Hong et al., 2024; Li et al., 2024b; Xu et al., 2024e;d; Zhang et al., 2024a; Han et al., 2024; Jiang & Wang, 2024; Xie et al., 2024a; Tang et al., 2024b; Tochilkin et al., 2024; Qian et al., 2024a; Szymanowicz et al., 2024; 2025a; Liang et al., 2025a; Szymanowicz et al., 2025b; Schwarz et al., 2025; Yang et al., 2025; Zhang et al., 2025a). However, in contrast to our method, these approaches remain limited to producing static 3D scenes. Other methods focus on specific scenes such as faces (Kirschstein et al., 2025). Concurrent work (Liang et al., 2025b; Xu et al., 2025) tackles real scenes but can not handle diverse generated scenes or large viewpoint changes.

4D generation. Recent years have seen rapid progress in 4D generation, i.e., dynamic 3D scene synthesis. Many approaches leverage input text prompts or images as guidance. Following the introduction of large-scale generative models for this task (Singer et al., 2023), subsequent works have made notable advances in both visual fidelity and motion quality (Ren et al., 2023; Ling et al., 2024a; Bahmani et al., 2024b; Zheng et al., 2024b; Gao et al., 2024a; Yang et al., 2024a; Jiang et al., 2024b; Miao et al., 2024; Li et al., 2024a; Zhang et al., 2024g; Yuan et al., 2024; Jiang et al., 2024a).

While text conditioning is widely used, alternative methods aim to lift 2D images or videos into dynamic 3D scenes (Ren et al., 2023; Zhao et al., 2023; Yin et al., 2023; Pan et al., 2024; Zheng et al., 2024b; Ling et al., 2024a; Gao et al., 2024a; Zeng et al., 2024b; Chu et al., 2024; Wu et al., 2024; Yang et al., 2024b; Wang et al., 2024c; Feng et al., 2024b; Sun et al., 2024c; Zhang et al., 2024c; Ren et al., 2024a; Lee et al., 2024b; Li et al., 2024c; Van Hoorick et al., 2024; Uzolas et al., 2024; Chai et al., 2024; Liang et al., 2024; Li et al., 2024d; Xie et al., 2024b; Li et al., 2025a; Wang et al., 2025e; Zhu et al., 2025; Zhou et al., 2025; Hu et al., 2025; Park et al., 2025). Other efforts explore incorporating physics priors into generation pipelines (Lin et al., 2024; Huang et al., 2024b; Zhang et al., 2024f; Kiray et al., 2025), or enhancing motion controllability with template-driven approaches (Zhang et al., 2024d; Sun et al., 2024b; Chen et al., 2024b). A parallel direction emphasizes compositional and interactive 4D generation (Bahmani et al., 2024a; Xu et al., 2024b; Cao et al., 2024; Yu et al., 2024a; Zeng et al., 2024a; Zhu et al., 2024).

Another strand of research extends 3D GAN frameworks into the 4D setting by training directly on 2D video data (Bahmani et al., 2023a; Xu et al., 2023). However, such methods are typically limited by small-scale, single-category datasets, which constrain generalization. Moreover, most existing approaches remain object-centric, often neglecting background elements. As a result, their overall visual fidelity lags behind the high photorealism achieved by recent video generation models, which our method leverages.

Camera-conditioned video models. Recent progress has focused on incorporating camera control into video diffusion models. The pioneering work MotionCtrl (Wang et al., 2024e) introduced camera conditioning by augmenting pre-trained video models (Chen et al., 2023a; Blattmann et al., 2023) with extrinsic matrices. Subsequent efforts (He et al., 2024a; Xu et al., 2024c; Kuang et al., 2024; Ju et al., 2025; He et al., 2025; Li et al., 2025c; Wang et al., 2025c; 2024d; Lei et al., 2025; Liu et al., 2025) enhanced conditioning by representing cameras using Plücker coordinates, while others use follow MotionCtrl to simply use poses flattened (Wang et al., 2025b). Another line of research (Hu et al., 2024; Xiao et al., 2024; Ling et al., 2024c; Hou et al., 2024) enabled camera motion control without introducing additional trainable parameters. Despite these advances, all of the above rely on U-Net-based architectures. Other approaches use low-rank approaches for camera control (Zhang et al., 2025b).

Several recent works continue to extend U-Net-based approaches for camera control (Zhao et al., 2024; Xu et al., 2024a; Zheng et al., 2024a; Zhang et al., 2024b; Yu et al., 2024b; Lei et al., 2024), while others begin exploring diffusion transformers. Cheong et al. (Cheong et al., 2024) propose one such transformer-based framework, though its scene and visual quality remain limited. DimensionX (Sun et al., 2024d) introduces joint space-time control in diffusion transformers, but relies on pre-defined, non-continuous camera trajectories. Other works (Huang et al., 2024a; Mai et al., 2025; Jiang et al., 2025b) instead explore pose and geometry estimation with a video DiT. CAT4D (Wu et al., 2025b) proposes a multi-view video diffusion model derived from fine-tuning a multi-view backbone. Large-

scale synthetic data generation further boosts performance, as demonstrated by SynCamMaster (Bai et al., 2025b) and ReCamMaster (Bai et al., 2025a), which achieve strong results on camera-controlled video generation using transformer-based architectures. Finally, 4DiM (Watson et al., 2024) trains a space-time diffusion model from scratch for novel view synthesis from a single input image. Another line of work edits camera movements of videos with underlying motion using camera retargeting (Seo et al., 2025; Ma et al., 2025). One popular direction has been to use a spatio-temporal cache, e.g., using point clouds or optical flow, to condition the camera-controlled video generation (Wu et al., 2025c; Ren et al., 2025; Wang et al., 2025d; Cao et al., 2025; YU et al., 2025; Jin et al., 2025; Xing et al., 2025; Yan et al., 2025; Li et al., 2025b; Gu et al., 2025).